Day 16 – Shell Scripting Basics

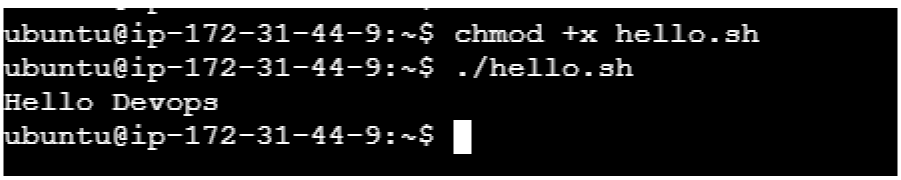
Task 1: Your First Script

shebang (#!/bin/bash) and why it matters- The shebang (#!/bin/bash) is a special line at the very beginning of a script file that indicates which interpreter should be used to execute the script.

hello.sh

#!/bin/bash

echo "Hello Devops"



Task 2: Variables

variables.sh

#!/bin/bash

Name="Parthavi"
Role="Cloud Engineer"

echo "Hello, I am $Name and my role is $Role"

```
ubuntu@ip-172-31-44-9:~$ ls -l variables.sh
-rw-rw-r-- 1 ubuntu ubuntu 97 Jun  1 04:20 variables.sh
ubuntu@ip-172-31-44-9:~$ chmod +x variables.sh
ubuntu@ip-172-31-44-9:~$ ls -l
total 32
drwxrwxr-x 2 berlin    heist-team 4096 May 31 17:20 app-logs
drwxrwxr-x 2 ubuntu    ubuntu     4096 May 31 17:48 bank-heist
-r--r--r-- 1 berlin    ubuntu        0 May 31 14:46 devops.txt
drwxrwxr-x 4 professor planners   4096 May 31 17:30 heist-project
-rwxrwxr-x 1 ubuntu    ubuntu       33 Jun  1 04:15 hello.sh
-rw-r----- 1 ubuntu    ubuntu       12 May 31 14:49 notes.txt
drwxr-xr-x 2 ubuntu    ubuntu     4096 May 31 15:06 project
-rw-rw-r-- 1 professor heist-team    0 May 31 17:14 project-config.yml
-rw-rw-r-- 1 professor heist-team    0 May 31 17:18 project-config1.yml
-rw-rw-r-- 1 ubuntu    ubuntu       13 May 31 14:52 script.sh
-rw-rw-r-- 1 ubuntu    heist-team    0 May 31 17:09 team-notes.txt
-rwxrwxr-x 1 ubuntu    ubuntu       97 Jun  1 04:20 variables.sh
ubuntu@ip-172-31-44-9:~$ ls -l variables.sh
-rwxrwxr-x 1 ubuntu ubuntu 97 Jun  1 04:20 variables.sh
ubuntu@ip-172-31-44-9:~$
```

```
ubuntu@ip-172-31-44-9:~$ ./variables.sh
Hello, I am Parthavi and my role is Cloud Engineer
ubuntu@ip-172-31-44-9:~$
```

1. Single Quotes '...'

Preserve the literal value of all characters inside the quotes.
Variables are not expanded or interpreted.
Special characters are treated as plain text.

Example:

NAME="Parthavi"
echo 'Hello, I am $NAME'

Output:
Hello, I am $NAME

Note: $NAME is printed literally, not replaced with the variable's value.

2. Double Quotes "..."

Allow variable expansion.
Variables inside double quotes are replaced with their current values.
You can include special characters that will be interpreted (like \n for newline).

Example:

```
    NAME="Parthavi"
echo "Hello, I am $NAME"
```

Output:

Hello, I am Parthavi

Note: $NAME is replaced with the value of the variable.

Task 3: User Input with read

```
#!/bin/bash

read -p "Name: " name
read -p "Tool: " tool

echo "Hello, I am $name and my favourite tool is $tool"
```

```
ubuntu@ip-172-31-44-9:~$ chmod +x greet.sh
\ubuntu@ip-172-31-44-9:~$ ./greet.sh
Name: parthavi
Tool: kubernetes
Hello, I am parthavi and my favourite tool is kubernetes
ubuntu@ip-172-31-44-9:~$
```

Task 4: If-Else Conditions

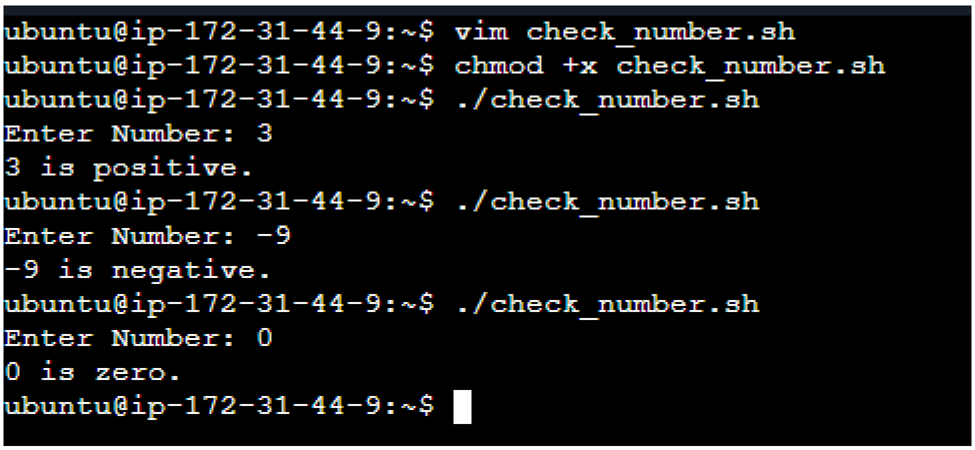
```
#!/bin/bash

read -p "Enter Number: " number

if [[ $number -gt 0 ]]; then
  echo "$number is positive."
elif [[ $number -lt 0 ]]; then
  echo "$number is negative."
else
  echo "$number is zero."
fi
```

Output:

ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 3
3 is positive.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: -9
-9 is negative.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 0
0 is zero.



```
ubuntu@ip-172-31-44-9:~$ vim check_number.sh
ubuntu@ip-172-31-44-9:~$ chmod +x check_number.sh
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 3
3 is positive.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: -9
-9 is negative.
ubuntu@ip-172-31-44-9:~$ ./check_number.sh
Enter Number: 0
0 is zero.
ubuntu@ip-172-31-44-9:~$
```

file_check.sh

File_check.sh

```
#!/bin/bash

read -p "Enter the filename: " filename

if [[ -f "$filename" ]]; then
   echo "The file '$filename' exists."
else
   echo "The file '$filename' does not exist."
fi
```

Output:

```
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: notes.txt
The file 'notes.txt' exists.
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: file.txt
The file 'file.txt' does not exist.
ubuntu@ip-172-31-44-9:~$ nano file_check.sh
```

```
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: notes.txt
The file 'notes.txt' exists.
ubuntu@ip-172-31-44-9:~$ ./file_check.sh
Enter the filename: file.txt
The file 'file.txt' does not exist.
ubuntu@ip-172-31-44-9:~$
```

Task 5: Combine It All

Server_check.sh

```
#!/bin/bash

# Store service name in a variable
service="sshd"

# Ask user for confirmation
read -p "Do you want to check the status? (y/n): " choice

if [ "$choice" = "y" ]; then
   if systemctl is-active --quiet "$service"; then
      echo "$service is active."
   else
      echo "$service is not active."
   fi

   # Show detailed status
   systemctl status "$service"
elif [ "$choice" = "n" ]; then
   echo "Skipped."
else
   echo "Invalid input. Please enter y or n."
fi
```

Output:

Do you want to check the status? (y/n): y sshd is active.

or

Do you want to check the status? (y/n): n

Skipped.

```
ubuntu@ip-172-31-44-9:~$ chmod +x server_check.sh
ubuntu@ip-172-31-44-9:~$ ./server_check.sh nginx
Get:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 Packages [1483 kB]
Get:6 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:7 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:8 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:9 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [103 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [33.3 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [23.6 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [968 B]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [74.6 kB]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [21.7 kB]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [39.7 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [628 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [174 kB]
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
package install hogaya
```

file_check.sh file_chec